

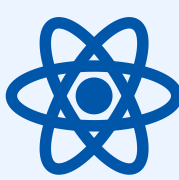
Learning by Launching: AI-Assisted Simulation and Two-Stage Water Rockets

A transdisciplinary service-learning project for primary-school science
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BACKGROUND / WHY IT MATTERS

Abstract mechanics can be difficult for young children to understand. Our team designed a Chinese-English STEM unit for **Primary 1-2 learners**, connecting rocket construction, interactive simulation and physical launches through a **Hong Kong-Jiangxi service-learning project**.



TRANSDISCIPLINARY

Physics × engineering
× computing × pedagogy



TRANSECTORAL

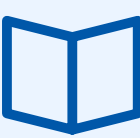
University × partner
school × community



TRANSCULTURAL

Hong Kong-Jiangxi
exchange and bilingual
teaching

OBJECTIVES



Understand

Connect air pressure, action-reaction and motion to an observable launch.



Investigate

Form a hypothesis → **test** → **reflect**; vary parameters and record observations.



Create

Build a rocket and connect simulated behaviour with hands-on experience.

KEY FINDINGS & IMPLICATIONS



≈95%

could explain basic
rocket principles

Teacher estimate from
classroom explanations;
no formal pre/post
assessment.



8

physical launch
attempts

Eight launches provided
opportunities to observe,
discuss and improve.



LAUNCH 3

of 8 launches
best observed run

Team reflection identified
the third launch as the
most successful attempt.

What we observed

Most learners could describe the basic launch principle. Construction, simulation and launch activities gave students several ways to connect pressure and force with visible motion.

What this suggests

The project combines science, engineering, computing and teaching in a service-learning setting. Its educational effectiveness needs systematic evaluation.

APPROACH & METHODOLOGY

1



LEARN

Explore
pressure and
forces.

2



BUILD

Assemble the
two-stage
rocket.

3



SIMULATE

Vary water,
pressure and
nozzle size.

4



LAUNCH

Observe and
record the
launch.

5



REFLECT

Compare,
discuss and
improve.



Rocket materials



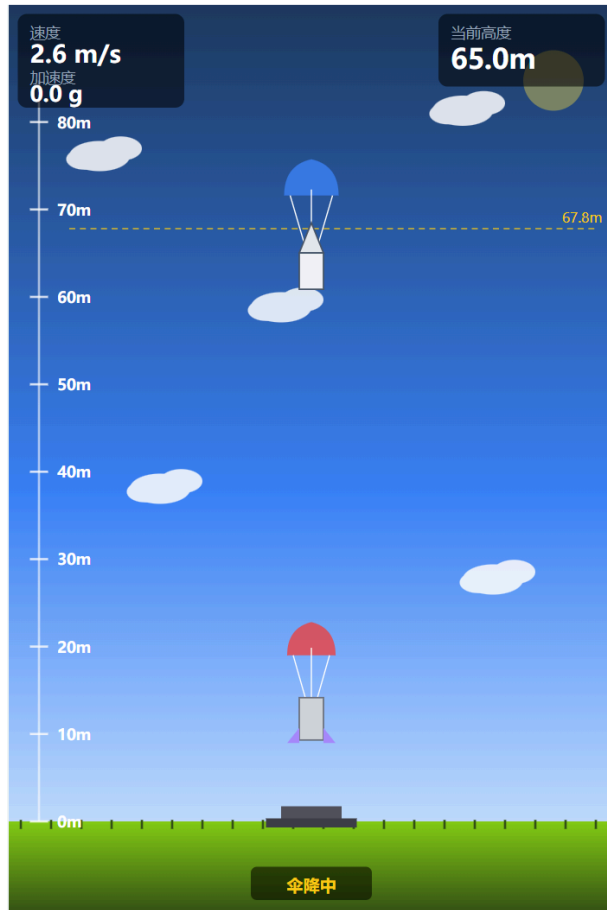
Building the rocket



Hands-on learning

SIMULATION AS A LEARNING TOOL

The simulator makes parameter changes and predicted trajectories visible before launch. Students can explore **cause and effect**, then compare the model with a physical experiment. Predictions support discussion; they are **not validated measurements**.



Actual simulator view

LEARNING AND COMPARISON CYCLE



INPUTS

Water volume
Air pressure
Nozzle diameter



SIMULATION

Predicted flight



LAUNCH

Physical
experiment



REFLECTION

Compare and
discuss

DEVELOPMENT THROUGH VIBE CODING



Design

Initial interface
and physics
framework



Refinement

Algorithm and
state-machine
refinement



Review

Check launch
sequence and
model behaviour



Deployment

Deployment
and iteration

Human review remains essential. AI-assisted development supported the simulator; instructors checked model behaviour and guided classroom use.

LIMITATIONS & NEXT STEPS

The 95% figure is a teacher estimate. Simulation accuracy and learning gains were not formally measured.

Next: add age-appropriate concept checks, structured launch records, simulation-experiment comparisons and a student design space.

REFERENCES

- Zacharia, Z., & Anderson, O. R. (2003). The effects of an interactive computer-based simulation prior to performing a laboratory inquiry-based experiment on students' conceptual understanding of physics. *American Journal of Physics*, 71(6), 618-629. doi:10.1119/1.1566427
- Wieman, C. E., Adams, W. K., & Perkins, K. K. (2008). PhET: Simulations That Enhance Learning. *Science*, 322(5902), 682-683. doi:10.1126/science.1161948

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Project & versions